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Contraception 79 (2009) 111 – 116

Contraception

Original research article

Insulin sensitivity and lipid metabolism with oral contraceptives containing chlormadinone acetate or desogestrel: a randomized trial☆,☆☆

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Received 1 August 2008; revised 2 September 2008; accepted 2 September 2008

Abstract

Background: Second-generation and third-generation oral contraceptives containing 30 mcg or more of ethinylestradiol (EE) decrease insulin sensitivity (SI). In this study, we investigated whether SI is decreased by contraceptives containing lower doses EE or by progestins with antiandrogenic properties.

Study Design: Twenty-eight young healthy women were randomly allocated to receive 20 mcg of EE and 150 mcg of desogestrel (DSG) (n=14) or 30 mcg of EE and 2 mg of chlormadinone acetate (CMA) (n=14) for 6 months. SI and glucose utilization independent of insulin (Sg) were investigated by the minimal model method. Lipid modifications were also analyzed.

Results: SI decreased with EE/DSG (7.09±1.4 vs. 4.30±0.91; p=.04; n=12), but not with EE/CMA (5.79±0.93 vs. 6.79±1.1; p=.48; n=12). SI modifications observed in the two groups were significantly different (−2.79±1.15 vs. 1.0±1.38; p=.05). Sg did not vary with either treatment. The response of C-peptide to glucose increased, but significantly so only with EE/CMA (p=.01). The C-peptide/insulin response increased with both EE/DSG (p=.05) and EE/CMA (p=.04). High-density lipoprotein (HDL) cholesterol (p=.02) and triglycerides (p=.02 and p=.01) increased in both groups, but HDL/low-density lipoprotein cholesterol (p=.02), apoprotein A1 (Apo-A1) (p=.04) and Apo-A1/apoprotein B (p=.048) increased significantly only with EE/CMA.

Conclusions: The present study confirms that DSG, even when associated with low EE dose, decreases SI. By contrast, EE/CMA does not deteriorate SI and induces a favorable lipid profile.

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Keywords: Contraception; Insulin; Metabolism; Progesterone; Chlormadinone acetate; Desogestrel

1. Introduction

Prevention of cardiovascular disease is a major goal of humans. Known cardiovascular risk factors often present together as features of metabolic syndrome [1–3]. The common alteration underlying the clinical manifestation of this syndrome is represented by decreased insulin sensitivity (SI) [1–3]. Insulin receptors [4] and SI vary throughout the menstrual cycle [5,6] as a consequence of gonadal steroid fluctuations. Optimal insulin receptors and SI are observed in the early midfollicular phase and decrease in the periovulatory and luteal phases with the additional increase in estrogen and progesterone [4–6]. Similarly, in postmenopausal women, elevated levels of estrogens or administration of progestin decreases SI [7,8]. Oral contraceptives inhibit, although incompletely [9], ovarian activity and create a new endocrine environment in which endogenous and exogenous hormones contribute to influence biological and metabolic functions. Oral contraceptives with ethinylestradiol (EE) in doses of 30 mcg or higher, which are associated with second-generation progestins (levonorgestrel), decrease peripheral insulin receptors [10] and induce subclinical abnormalities in carbohydrate metabolism [11–13]. Contraceptives with third-generation progestins [desogestrel (DSG) and gestodene] are considered more neutral [14], but preparations with

☆ This study received unrestricted funds from the Department of Obstetrics and Gynecology, University of Modena.

☆☆ The present trial was registered with the European Clinical Trial Database (no. 2007-007931-25).

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0010-7824/$ – see front matter © 2009 Elsevier Inc. All rights reserved.
doi:10.1016/j.contraception.2008.09.002

an EE dose of 30 mcg or higher that are associated with gestodene [15] or DSG [13,16] still decrease SI. There are no available data for compounds containing doses of EE lower than 30 mcg.

Similarly, in healthy women, there are no available data for compounds containing progestins devoid of androgenic properties. Indeed, both second-generation and third-generation progestins are 19-nor testosterone derivatives [17], and several pieces of evidence support the possibility that the residual androgenic activity of these molecules — greater for second-generation progestins than for third-generation progestins — may play a role in reducing SI. SI is decreased by androgen supplementation [18–21], is reduced in hyperandrogenic women with polycystic ovary syndrome [22,23] and, in the latter, can be improved by abolition of hyperandrogenism [24,25] or by administration of antiandrogens [16,26]. Chlormadinone acetate (CMA) is a 17-OH progesterone derivative molecule with peripheral antiandrogenic properties [27]. Initially used as a progestin-only contraceptive [28], CMA is now proposed in combination with 30 mcg of EE as an oral contraceptive for all women [29]. In the present study, we investigated the modification in SI induced by the administration of two contraceptives: one containing a third-generation progestin (DSG) but a low 20-mcg dose of EE, and the other containing the usual 30-mcg dose of EE but a progestin dose devoid of androgenic properties (CMA). Modifications in lipid profile were evaluated as well.

2. Materials and methods

Between March and June 2007, lean [body mass index (BMI) below 25] healthy women aged 18–35 years were enrolled by a physician from our hospital's outpatient service for contraception. The enrolled women had normal menstrual cycles in the previous 12 months, with a cycle length ranging between 28 and 31 days, and had been free from oral contraceptives or medicines that could possibly influence glucose and lipid metabolism for the 12 months preceding the study. Among the 48 women screened, 39 women were eligible, and 28 gave their informed consent to the study, which had been previously approved by our local ethics committee and the Institutional Review Board. The primary outcome of the study was modification in SI induced by each treatment. Secondary outcomes were modifications in glucose utilization independent of insulin (Sg), β-cell response to glucose, insulin clearance, and modifications in lipoproteins and liver enzymes.

In each woman, SI and Sg were investigated with a frequently sampled intravenous glucose tolerance test (FSIGT) associated with the minimal model method [30–32]. Investigation was performed at baseline during the early follicular phase (4–7 days after spontaneous menstruation). After the baseline evaluation, each woman was allocated to receive for 6 months either (a) a monophasic oral contraceptive containing 20 mcg of EE plus 150 mcg of DSG (Mercilon; Organon, Oss, Holland) (n=14); or (b) a monophasic oral contraceptive containing 30 mcg of EE and 2 mg of CMA (Belara; Grunenthal-Formenti, Milan, Italy) (n=14). Allocation to treatment was randomized following a computer-generated list of randomization. The list was concealed, and an independent physician performed the allocation. During the 6-month study, women were requested not to modify their lifestyle or dietary habit. Investigations were repeated during the last 7 days of the last month of treatment.

2.1. FSIGT evaluation

Each woman was hospitalized at 0700 h, following a 12-h overnight fast and a 3-day diet containing at least 200 g/day carbohydrates. Women were maintained on bed rest. Two polyethylene catheters were placed in two antecubital veins and were kept patent by a slow infusion of saline solution. One catheter was used for intravenous glucose or insulin administration, and the other catheter was used for blood collection. At 0900 h, glucose (0.3 g/kg) was injected intravenously over 1 min and was followed 20 min later by an intravenous insulin bolus (0.03 U/kg). Samples of arterialized blood were collected at −5, −1, 2, 4, 8, 20, 22, 30, 40, 60, 70, 100 and 180 min after glucose load [32].

2.2. Assays

Blood samples were collected into tubes placed on ice and immediately centrifuged. An aliquot of serum was immediately tested for glucose levels, whereas another aliquot was immediately frozen at −25°C until assayed. Glucose was determined by the glucose oxidase method. Insulin levels were assayed in all samples (in duplicate) by radioimmunoassay (RIA) methods using commercial kits (Biodata; Guidonia Montecelio, Rome, Italy). Intrassay and interassay coefficients of variation of the assay were 6.2% and 7%, respectively, and sensitivity was 14.35 pmol/L. C-peptide levels were analyzed in the samples collected in the first 20 min of FSIGT. Assays were performed in duplicate using commercial RIA kits (Biodata; Guidonia Montecelio). Intraassay and interassay coefficients of variation were 3.2% and 8.5%, respectively, and sensitivity was 33.1 pmol/L. Circulating levels of luteinizing hormone (LH), follicle-stimulating hormone (FSH), estradiol and testosterone were also analyzed with RIA in the first sample of FSIGT. To avoid interassay variability, samples of each subject were analyzed altogether in the same assay.

Liver function and lipid metabolism were evaluated by assays of liver enzymes such as glutamic–oxalacetic transaminase (GOT), glutamic–pyruvic transaminase (GPT) and γ-glutamine transferase (γGT), and assays of total cholesterol, high-density lipoprotein (HDL) cholesterol, low-density lipoprotein (LDL) cholesterol, triglycerides, apoprotein A1 (Apo-A1) and apoprotein B (Apo-B) in the first sample of FSIGT. Commonly used laboratory

Table 1
Mean (±SE) age, BMI, liver enzymes and endocrine parameters observed in healthy young women before and after 6 months of treatment with EE+DSG (EE/DSG; *n*=12) or EE+CMA (EE/CMA; *n*=12)

	EE/DSG			EE/CMA		
	Before	After	p	Before	After	p
AGE (years)	27.8±1.3			28.1±1.1		
BMI (kg/m^2)	23.0±0.9	23.6±1.0	.08	23.1±1.0	23.3±0.9	.32
GOT (U/L)	16.5±0.8	15.6±0.8	.39	18.5±1.7	17.4±1.7	.21
GPT (U/L)	16.5±2.0	16.1±2.3	.83	16.7±2.4	16.2±2.5	.85
γGT (U/L)	10.8±1.5	12.0±2.1	.32	12.0±1.4	9.9±1.7	.15
LH (IU/L)	5.9±1.6	1.0±0.3	.018	5.2±0.7	1.8±0.65	.001
FSH (IU/L)	8.2±1.4	1.5±0.6	.001	7.3±1.3	1.9±0.4	.001
E (pmol/L)	250.7±93.9	182.5±99.8	.056	232.9±51.4	78.2±12.1	.011
T (nmol/L)	188.9±64.2	155.6±62.4	.13	150.8±79.0	128.8±22.4	.36

E, estradiol; T, testosterone.

techniques were used to measure liver enzymes. Plasma total cholesterol and triglycerides were measured by enzymatic methods (Olympus), while HDL-cholesterol was determined after precipitation with PEG 6000. LDL-cholesterol levels were calculated by the Friedewald formula. Apo-A1 and Apo-B were determined by immunonephelometry on a Behring nephelometer analyzer (Behringwerke Marburg, Germany).

2.3. Statistical analyses

Assuming that the difference induced by treatment was equal to 1 SD of the difference and by setting type I error at 0.05 and type II error at 0.20, eight subjects were sufficient in each group to detect a statistically significant modification in the FSIGT investigation. The responses of glucose, insulin and C-peptide observed during the first 20 min of FSIGT were reported as area under the curve (AUC), calculated by the trapezoid method and expressed in arbitrary units (nmol/L min or pmol/L min). To have an index of hepatic insulin clearance, the C-peptide/insulin ratio of integrated responses to glucose challenges was calculated [16,25]. Glucose and insulin values obtained during the FSIGT were used to calculate SI [30,32] and Sg, using a computerized algorithm (MIN-MOD). (SI is expressed in units×10^{-4} min^{-1}×μU/mL, and Sg is expressed in units×10^{-4} min^{-1}.)

Data were analyzed by one of the authors (A.C.), who was blinded to the specific treatment of each group, as the data of two groups. Analysis was performed with the statistical package Statview 5.0.1 for Apple MacIntosh (SAS Institute Inc., 1998). Within-group comparison was performed with *t* test for paired data. Two-way analysis of variance for repeated measures (Treatment×Time, with subjects as replicates) and Student's *t* test were used to evaluate differences between the two treatment groups. The results are expressed as mean±standard error (SE).

3. Results

No woman withdrew from the study for relevant side effects. Of the 28 women enrolled, four women (two for each group) stopped contraceptive use for personal reasons (*n*=1, desire for pregnancy; *n*=1, no need for contraception), were lost to follow-up (*n*=1) or did not want to repeat the investigation (*n*=1). Accordingly, analyses were performed in the 24 women who finished the study (12 for each group). Basal conditions and modifications of BMI, circulating

Table 2
Mean (±SE) metabolic parameters observed in healthy young women before and after 6 months of treatment with EE+DSG (EE/DSG; *n*=12) or EE+CMA (EE/CMA; *n*=12)

	EE/DSG			EE/CMA		
	Before	After	p	Before	After	p
Insulin (pmol/L)	71.5±8.8	86.5±11.3	.17	66.3±12.9	78.8±9.7	.11
Glucose (mmol/L)	4.7±0.1	4.6±0.8	.34	4.5±0.1	4.5±0.1	.88
C-peptide (pmol/L)	964.9±345.3	1261.1±404.5	.09	739.6±110.5	1059.5±100.3	.01
Cholesterol (mmol/L)	4.24±0.33	4.70±0.36	.02	4.12±0.21	4.45±0.28	.02
HDL-C (mmol/L)	1.40±0.10	1.66±0.15	.02	1.60±0.11	1.92±0.10	.02
LDL-C (mmol/L)	2.49±0.29	2.56±0.35	.66	2.29±0.22	2.09±0.25	.16
HDL/LDL	0.65±0.1	0.85±0.2	.12	0.78±0.1	1.1+0.2	.045
Triglyceride (mmol/L)	0.75±0.06	1.03±0.14	.01	0.63±0.08	0.93±0.08	.02
Apo-A1 (mg/dL)	126.1±7.1	132.1±18.3	.78	161.7±8.8	196.7±14.0	.04
Apo-B (mg/dL)	73.4±8.1	80.8±8.8	.29	80.3±7.5	81.0±7.3	.78
Apo-A1/Apo-B	1.89±0.26	2.0±0.46	.68	2.15±0.18	2.59±0.27	.048

levels of glucose, insulin, C-peptide, LH, FSH, estradiol, testosterone, liver enzymes and lipids during treatment are reported in Tables 1 and 2. BMI and liver enzymes were not affected by either contraceptive. Both contraceptives decreased levels of FSH, LH and estradiol. Levels of testosterone did not vary.

3.1. EE/DSG

In women receiving EE/DSG, the response of glucose to FSIGT (123.0±5.2 vs. 125.0±3.3 AUC; p=.69) was not modified. The responses of insulin (3962.4±496.3 vs. 5086.4±766.5 AUC; p=.10) and C-peptide (32,720.3±11,623.8 vs. 45,906.4±11,884.4 AUC; p=.16) tended to increase, although not significantly. The response of the C-peptide/insulin ratio to FSIGT significantly increased (9.24±5.2 vs.11.8±4.9; p=.05) (Fig. 1).

The minimal model evaluation of the FSIGT data showed no modification in Sg (0.032±0.003 vs. 0.024±0.003; p=.18) and a significant decrease in SI (7.09±1.4 vs. 4.30±0.91; p=.04) (Fig. 2).

Total cholesterol increased (p=.02). Lipoprotein analysis revealed a significant increase in HDL-cholesterol (p=.02) along with triglycerides (p=.01), but no modification in LDL-cholesterol. The HDL/LDL ratio, Apo-A1, Apo-B or Apo-A1/Apo-B ratio was not significantly modified (Table 2).

3.2. EE/CMA

In women receiving EE/CMA, integrated FSIGT responses of glucose (130.3±6.1 vs. 123.9±5.1 AUC; p=.39) and insulin (3220.8±743.6 vs. 2991.9±229.2 AUC; p=.72) did not vary significantly, while those of C-peptide (21,467.1±6459.9 vs. 39,108.4±6453.4 AUC; p=.01) and the C-peptide/insulin ratio (9.34±2.1 vs. 13.9±2.5; p=.04) significantly increased (Fig. 1). The minimal model evaluation of the FSIGT results showed no

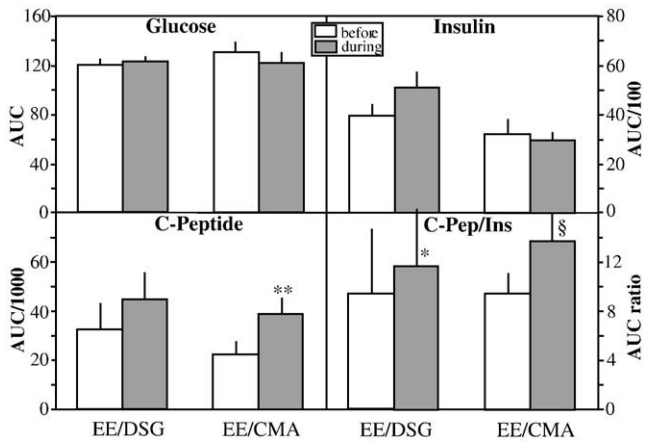

Fig. 1. Mean (±SE) glucose, insulin, C-peptide and C-peptide/insulin integrated response to intravenous glucose administration, as observed in healthy young women before and after 6 months of treatment with EE+DSG (EE/DSG; *n*=12) or EE+CMA (EE/CMA; *n*=12). §p=.04; *p=.05; ** p=.01 (versus corresponding before values).

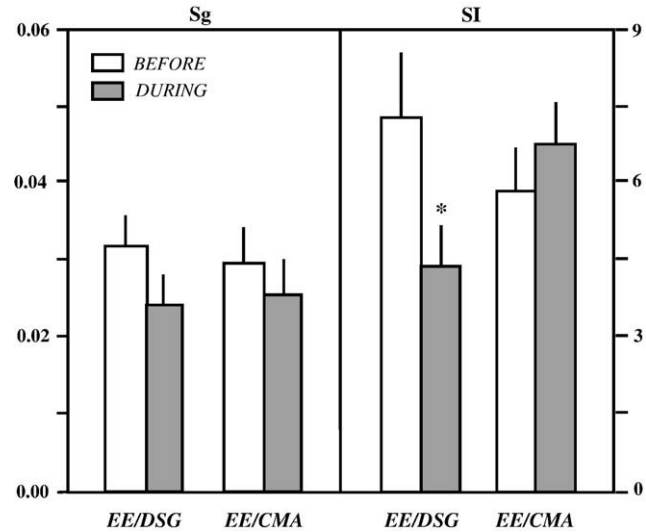

Fig. 2. Mean (±SE) glucose-dependent glucose utilization (Sg; units×10^{-4} min^{-1}) (left) and peripheral SI (units×10^{-4} min^{-1}×μ/mL) (right) observed in healthy young women before and after 6 months of treatment with EE+DSG (EE/DSG; *n*=12) or EE+CMA (EE/CMA; *n*=12). *p=.04 (versus corresponding before values).

significant modification in Sg (0.029±0.003 vs. 0.026±0.003; p=.28) and SI (5.79±0.93 vs. 6.79±1.1; p=.48) (Fig. 2).

Total cholesterol increased (p=.02). Lipoprotein analysis revealed a significant increase in HDL-cholesterol (p=.01) and triglycerides (p=.02), but no modification in LDL-cholesterol. Apo-B did not vary, while a significant increase was observed in the HDL/LDL ratio (p=.04), Apo-A1 (p=.04) and Apo-A1/Apo-B ratio (p=.05) (Table 2).

3.3. EE/DSG versus EE/CMA

No difference in baseline parameters was observed between the two groups of women. A significant interaction that decreased with EE/DSG (−2.79±1.15) and tended to increase with EE/CMA (1.0±1.38) was observed for SI (p=.05).

No significant difference in hormone and metabolic modifications was observed between the two groups, although only EE/CMA induced a significant increase in the HDL/LDL ratio, Apo-A1 and Apo-A1/Apo-B ratio.

4. Discussion

In the present study, we report the effects of two oral contraceptives on the SI of young healthy women. The results show that the two compounds act differently. EE/DSG significantly decreases SI, while EE/CMA does not modify SI.

The data indicate that the reduction of EE to 20 mcg does not abolish the negative impact of EE and DSG on SI, which has been previously described with compounds containing higher EE doses (40 and 30 mcg) [13,16]. Likely, DSG,

independently of EE, negatively affects SI. Indeed, SI is decreased also by the subdermal implant of etonorgestrel, the active DSG metabolite [33].

We hypothesized that the decrease in SI associated with the administration of second-generation and third-generation oral contraceptives, including DSG, could be related to the androgenic properties of these progestins [17], with lower detrimental effect probably being induced by molecules with antiandrogenic properties. The data obtained with EE/CMA seem to confirm this possibility. CMA had antiandrogenic properties [27], did not decrease SI and even tended to increase SI.

Comparison of SI modifications induced by EE/DSG and EE/CMA revealed, as a secondary analysis, a statistically significant difference that emphasizes the importance of selecting the right contraceptive for optimal SI maintenance.

Other parameters of glucose metabolism were similarly affected by the two contraceptives. The pancreatic response to glucose (C-peptide) was increased by both associations, but significantly so only by EE/CMA. Insulin clearance was also increased significantly by both contraceptives. The mechanism mediating the increase in pancreatic β-cell response to glucose and the increase in insulin clearance is not clear, but it resembles the effect induced by the administration of oral estrogens to postmenopausal women [34]. Accordingly, this effect, which is beneficial for carbohydrate metabolism [35], may be the consequence of the estrogenic balance of these compounds (greater for the EE/CMA association).

Data on lipids seem to confirm this view. It is known that estrogen-balanced oral contraceptives increase HDL-cholesterol and triglycerides and decrease LDL-cholesterol [36]. The opposite occurs during the administration of androgen-balanced oral contraceptives [36]. Both EE/CMA and EE/DSG elevated HDL-cholesterol and triglycerides, indicating an estrogen-balanced equilibrium, but only the effect of EE/CMA was strong enough to significantly increase the HDL/LDL ratio, Apo-A1 and Apo-A1/Apo-B ratio. The modification induced by EE/DSG was less pronounced because of lower estrogenic balance. Data on lipids obtained with EE/DSG are in line with those previously reported in the literature [37–40]. Results obtained with EE/CMA support observations previously published in condensed form [41] and, although limited, indicate a favorable lipid profile.

A limited number of subjects were included in this study, and the results should be considered preliminary. Nevertheless, the method used to evaluate SI was rather reliable and had been used in previous studies with groups with orders of magnitude similar to ours [4–8]. We were clearly able to show the negative effect of EE/DSG on SI. No significant modification in SI was observed with EE/CMA. Power analysis indicates that 38 subjects would have been necessary to show a possible positive effect of this association. We were also able to show positive modifications of lipids, in particular of HDL/LDL ratio and Apo-A1/Apo-B, following EE/CMA. The modification observed with EE/DSG showed the same trend but did not reach statistical significance. In this case, the HDL/LDL ratio would have required only a few more cases to reach statistical significance (n=16), while the Apo-A1/Apo-B ratio would have required not less than 53 cases.

Between-groups comparison was a secondary analysis, and the number of subjects analyzed allowed the documentation of a significant intergroup difference in SI modification. No other difference in glucose or lipid metabolism would have reached statistical significance unless groups included a number of subjects ranging from 50 to 800.

Contraception is increasingly being used for longer periods and at ages when metabolic imbalances become manifest. It is necessary that new contraceptives, by fulfilling these increasing demands, do not increase the risk of diseases, including cardiovascular diseases. The present study reports novel and encouraging findings for the EE/CMA association. EE/CMA does not deteriorate SI and induces a favorable lipid profile. These results—although necessitating strengthening by other experiences—are reassuring for the long-term use of this association.

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